

Assignment 1: Which Sleep-Wake States Are Induced by Optogenetic Activation? A Coding-Based Exploration (15%)

You are a new trainee in a lab studying the neural circuits that regulate sleep and wakefulness. You are recording activity from three exploratory brain regions and aim to determine which sleep-wake states they regulate. To investigate this, you will use optogenetics to activate each region and analyze physiological features (such as cortical and muscle activity) of these laser-induced states. Based on these features, you will classify the sleep-wake state induced by activation of each region (Peever & Fuller, 2017).

You know that mice have three main sleep-wake states: wakefulness, non-rapid eye movement (NREM) sleep, and rapid eye movement (REM) sleep (**fig 1**).

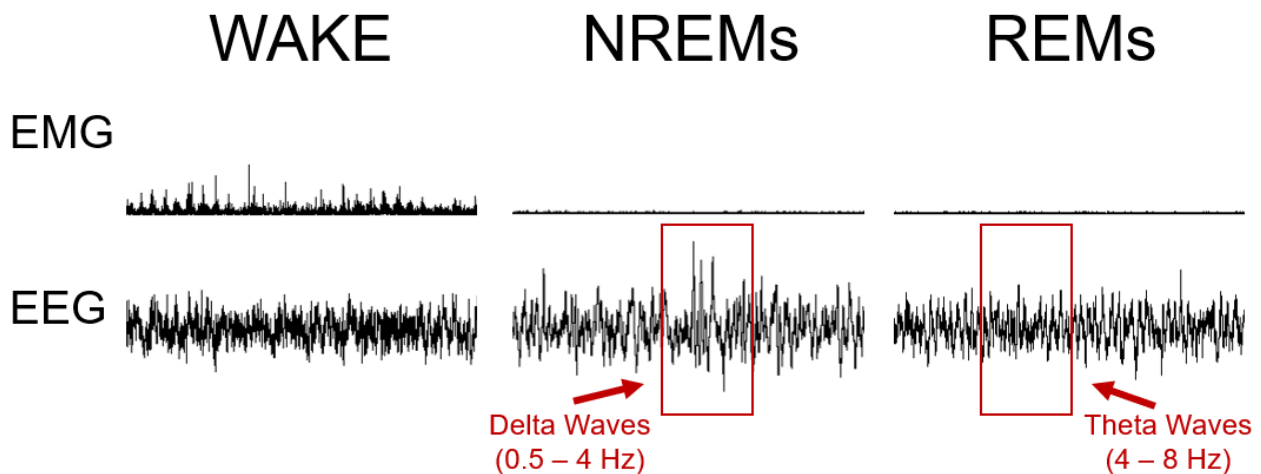


Fig 1: Example trace of EEG and EMG recording across different sleep-wake states

You can easily distinguish these states based on cortical activity recorded by electroencephalogram (EEG) and muscle activity recorded by electromyography (EMG).

EEG activity can be divided into the following frequency bands:

- Delta (0.5 – 4 Hz)
- Theta (4-8 Hz)
- Alpha (8 – 14 Hz)
- Beta (14-30 Hz)
- Gamma (30 – 100 Hz)

In contrast, EMG activity reflects movement. Higher EMG activity is generally associated with wakefulness, whereas lower EMG activity is sleep.

You searched the scientific literature on PubMed and identified several physiological features that can help you distinguish among these states:

1. NREM sleep is characterized by high delta power relative to wakefulness and REM sleep.
2. Both NREM and REM sleep are associated with little or no muscle activity.
3. Wakefulness is characterized by high muscle activity (a lot of movement).
4. Wakefulness typically exhibits prominent theta and gamma power.
5. REM sleep is characterized by prominent theta power.
6. Gamma power is lower during NREM and REM sleep than during wakefulness.

To manipulate your three exploratory regions and test which state they generate, you injected these mice with an excitatory opsin and implanted an optic fiber in the same region to activate these cells (**see Lecture 2 - Optogenetics**). Additionally, you implanted an electrophysiology apparatus to record the mouse's brain waves and muscle activity (**see Lecture 1 - Electrophysiology**).

You realize that identifying these states manually would be very time-consuming, so you decide to write a Python program that can automatically classify sleep-wake states across all your experiments.

Part 1: Cortical Activity (EEG) Analysis

In your code, make sure that your functions perform the following:

- Take the raw EEG data from each optogenetic activation trial as input (2-3 trials per mouse)
- Convert the raw EEG signal from the time domain to the frequency domain using an appropriate mathematical method (sampling rate = 1024 Hz)
- Separate the EEG power values into the frequency bands that you are previously researched
- Calculate the power within each frequency band for every trial and then average the power across the three trials separately for each brain region
- Return the total power for frequency bands of interest for each trial

Part 2: Muscle Activity (EMG) Analysis

In your code, make sure that your functions perform the following:

- Take the raw EMG data from each optogenetic activation trial as input (three trials per brain region)

- Normalize the EMG data so that muscle activity can be compared across trials and animals
- Calculate the average muscle activity for each trial
- Average the muscle activity across the three trials separately for each brain region
- Return the average muscle activity for each brain region

Part 3: Identifying functions of each brain region

Lastly, you will graph your results for each brain region and interpret them. Based on the physiological features shown in your graphs, determine whether activation of each region promotes wakefulness, NREM sleep, or REM sleep.

Part 4: Identifying the Activity Patterns of the Cells of Interest

You recorded neuronal signals from one of your brain regions of interest using fiber photometry. Unfortunately, you forgot to label the recordings! You realize that by determining when the cells are most active, you may be able to identify the mislabeled brain region.

You decide to write Python code to calculate:

- The average neuronal activity during each sleep-wake state
- The number of signal peaks (transients) during each state

Describe the activity pattern of these cells. Were the results consistent with your expectations? Based on their activity across the different sleep–wake states, which brain region do you think the recording corresponds to?

Data is collected and provided by Dr. Russell Luke and Irina Alymova in the Peever-Fraigne lab.

References:

Peever, J., Fuller, P.M. (2017). The biology of REM sleep. *Curr Biol* 20, 1237–1248.

Assignment Rubric

Coding Plan (1% of your final grade): Prepare a brief plan outlining how you will approach the assignment. The plan does not need to contain complete or working code. It should demonstrate that you have considered how to organize your analysis before beginning the implementation.

Code Commentary and Readability (2% of your final grade): clear variable names, explanatory comments and structure of the code.

Completion Grade (5% of your final grade): To help you with your final projects, weekly submissions will count toward your completion grade for this assignment. This means you'll receive marks for participating and submitting your work on time, regardless of whether your code is fully working. The goal of these mini-assignments is to help you build your final code bit by bit, following along with the tutorials. Each submission is worth 2% of your final assignment grade. The submission schedule is attached below.

Code Correctness (6% of your final grade): accurate logic, functions and plots; code works as intended and return correct values; proper stats and code execution

Interpretation (1% of your final grade): you should explain clearly what your results mean. This includes interpretation of the final results of your assignments. Make sure to be thoughtful and specific in your explanation (for example, explain why your results support or do not support your hypothesis). One short paragraph is sufficient.

Mini-Assignments Deadline Schedule:

Coding Plan: September 21st, 2026 at 11:59 pm

Assignment 1.1: September 28th, 2026 at 11:59 pm

Assignment 1.2: October 5th, 2026 at 11:59 pm

Assignment 1.3: October 12th, 2026 at 11:59 pm

Final Assignment Deadline: October 21st, 2026 at 11:59 pm